End-Term Examination
(SUBJECTIVE TYPE)(OffLine)
Course Name:<B.Tech>, Semester:<I>
(December, 2025)

Subject Code: BAS 102	Subject: Applied Physics
Time :3 Hours	Maximum Marks :60

Note: Q1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		(2.5*8=20)	CO Mapping
	a) Unpolarized light falls on the two polarizing sheets placed in a row. What must be the angle between the characteristic directions of the sheets if the intensity of the transmitted light through the second polarizing sheet is one third intensity of the incident beam?		CO1
	b) Deduce the missing orders for a double slit Fraunhofer diffraction pattern, if the slit widths are 0.16 mm and they are 0.8mm apart.		CO1
	c) Find whether the given vector field is irrotational in nature or not. $\vec{A}= (x^2y)\, i + (2xy)\, j + (z^2)\, k$		CO2
	d) State Stokes and Divergence theorem.		CO2
	e) What is pulse dispersion in optical fiber?		CO4
	f) An optical fiber has the refractive index of the core and cladding as 1.545 and 1.496 respectively. Calculate its numerical aperture and acceptance angle.		CO4
	g) What should be the kinetic energy of a neutron in eV so that its associated de Broglie wavelength is 1.4×10^{-10} m. mass of neutron= 1.675×10^{-27} kg		CO3
	h) What are the essential properties that a wave function must satisfy in quantum mechanics?		CO3
	UNIT I		CO Mapping
Q2	(a)Discuss the phenomenon of Fraunhofer diffraction in N-slit arrangement. Derive the expression for the intensity distribution and explain the conditions for the principal maxima, secondary maxima, and minima in the diffraction pattern. (b) Calculate the angles at which first dark band and the next bright band are formed in the Fraunhofer diffraction of a slit 0.3mm wide, wavelength of light is 5890 Å.	(7, 3)	CO1
Q3	(a) Derive Fresnel's Diffraction at a straight edge for a line source. Show the graphical representation of the diffraction pattern in the illuminated and the geometrical shadow regions. (b) Calculate the thickness of a quarter wave plate given that Refractive index of ordinary ray (μ_o) = 1.544 and Refractive index of extraordinary ray (μ_e) = 1.553 and wavelength = 5000 Å	(7,3)	CO1

	UNIT II		**CO Mapping**
Q4	(a) Derive 3rd and 4th Maxwell's equations starting from their fundamental laws and state their physical significance. (b) A plane electromagnetic wave has the magnetic field given by $\overrightarrow{B} = B_o \sin[(x+y)\frac{k}{\sqrt{2}} + \omega t]\,\hat{k}$ Where k is wave number Find (i) the electric field (ii) Poynting vector	**(5,5)**	**CO2**
Q5	(a) Using Maxwell's equations, derive and solve the wave equations for the transverse electric and magnetic fields in a conducting medium. Find the impedance offered by the conducting medium. (b) An electromagnetic wave of frequency f = 1 MHz is incident on a good conductor with conductivity 5.8×10^7 S/m. Calculate the skin depth of the wave in the conductor. (Consider $\mu_o = 4\pi \times 10^{-7}$ H/m, $\mu_r = 1$)	**(7, 3)**	**CO2**
	UNIT III		**CO Mapping**
Q6	(a) Derive the normalized wave function and energy eigenvalues for a particle confined in a one-dimensional infinite potential well of length L. Using the obtained wave functions, sketch the wave functions for the first two energy states n=1,2. (b) A particle of mass $m=9.11\times10^{-31}$ kg is confined in a one-dimensional infinite potential box of length $L=1.0\times10^{-10}$ m. calculate the energy of the 3rd and 4th excited state of the particle.	**(5, 5)**	**CO3**
Q7	(a) Define Heisenberg's uncertainty principle and explain its significance. Apply it to prove the non-existence of electrons inside the nucleus of an atom. (b) A particle in one dimension is represented by the given wavefunction $\psi(x) = \begin{cases} 0 & , x<a \\ A\,(1+\cos(\frac{\pi x}{a})) & , -a\le x\le a \\ 0 & , x>a \end{cases}$ Find the constant A, so that $\psi(x)$ is normalized.	**(7, 3)**	**CO3**
	UNIT IV		**CO Mapping**
Q8	a. Explain the working principle of a laser and, derive the Einstein coefficients of spontaneous and stimulated emission. b. Explain the working of He-Ne laser with its energy level diagram. Also discuss its applications.	**(5,5)**	**CO4**
Q9	(a) Explain the schematic diagram of an optical fiber communication system. Also discuss its major advantages. (b) Differentiate between step index and graded index optical fiber.	**(7,3)**	**CO4**